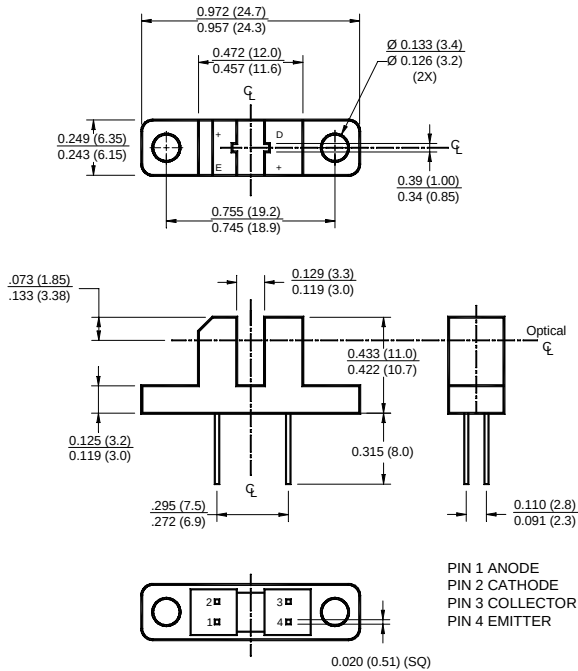


H21A1 / H21A2 / H21A3

PHOTOTRANSISTOR

OPTICAL INTERRUPTER SWITCH

PACKAGE DIMENSIONS

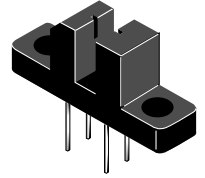


NOTES:

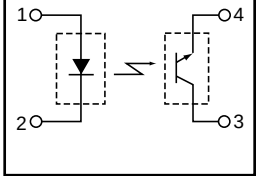
- Dimensions for all drawings are in inches (mm).
- Tolerance of $\pm .010$ (.25) on all non-nominal dimensions unless otherwise specified.

DESCRIPTION

The H21A1, H21A2 and H21A3 consist of a gallium arsenide infrared emitting diode coupled with a silicon phototransistor in a plastic housing. The packaging system is designed to optimize the mechanical resolution, coupling efficiency, ambient light rejection, cost and reliability. The gap in the housing provides a means of interrupting the signal with an opaque material, switching the output from an "ON" to an "OFF" state.



SCHEMATIC



FEATURES

- Opaque housing
- Low cost
- .035" apertures
- High $I_{C(ON)}$

- Derate power dissipation linearly 1.33 mW/°C above 25°C.
- RMA flux is recommended.
- Methanol or isopropyl alcohols are recommended as cleaning agents.
- Soldering iron tip 1/16" (1.6mm) minimum from housing.

ABSOLUTE MAXIMUM RATINGS ($T_A = 25^\circ\text{C}$ unless otherwise specified)

Parameter	Symbol	Rating	Unit
Operating Temperature	T_{OPR}	-55 to +100	°C
Storage Temperature	T_{STG}	-55 to +100	°C
Soldering Temperature (Iron) ^(2,3 and 4)	T_{SOL-I}	240 for 5 sec	°C
Soldering Temperature (Flow) ^(2 and 3)	T_{SOL-F}	260 for 10 sec	°C
INPUT (EMITTER)			
Continuous Forward Current	I_F	50	mA
Reverse Voltage	V_R	6	V
Power Dissipation ⁽¹⁾	P_D	100	mW
OUTPUT (SENSOR)			
Collector to Emitter Voltage	V_{CEO}	30	V
Emitter to Collector Voltage	V_{ECO}	4.5	V
Collector Current	I_C	20	mA
Power Dissipation ($T_C = 25^\circ\text{C}$) ⁽¹⁾	P_D	150	mW

H21A1 / H21A2 / H21A3

PHOTOTRANSISTOR

OPTICAL INTERRUPTER SWITCH

ELECTRICAL / OPTICAL CHARACTERISTICS (T _A =25°C)(All measurements made under pulse condition)							
PARAMETER	TEST CONDITIONS	SYMBOL	DEVICES	MIN	TYP	MAX	UNITS
INPUT (EMITTER)							
Forward Voltage	I _F = 60 mA	V _F	All	—	—	1.7	V
Reverse Breakdown Voltage	I _R = 10 μA	V _R	All	6.0	—	—	V
Reverse Leakage Current	V _R = 3 V	I _R	All	—	—	1.0	μA
OUTPUT (SENSOR)							
Emitter to Collector Breakdown	I _F = 100 μA, Ee = 0	BV _{ECO}	All	6.0	—	—	V
Collector to Emitter Breakdown	I _C = 1 mA, Ee = 0	BV _{CEO}	All	30	—	—	V
Collector to Emitter Leakage	V _{CE} = 25 V, Ee = 0	I _{CEO}	All	—	—	100	nA
COUPLED							
On-State Collector Current	I _F = 5 mA, V _{CE} = 5 V	I _{C(ON)}	H21A1	0.15	—	—	mA
			H21A2	0.30	—	—	
			H21A3	0.60	—	—	
	I _F = 20 mA, V _{CE} = 5 V		H21A1	1.0	—	—	
			H21A2	2.0	—	—	
			H21A3	4.0	—	—	
	I _F = 30 mA, V _{CE} = 5 V		H21A1	1.9	—	—	
			H21A2	3.0	—	—	
			H21A3	5.5	—	—	
Saturation Voltage	I _F = 20 mA, I _C = 1.8 mA	V _{CE(SAT)}	H21A2/3	—	—	0.40	V
	I _F = 30 mA, I _C = 1.8 mA		H21A1	—	—	0.40	V
Turn-On Time	I _F = 30 mA, V _{CC} = 5 V, R _L = 2.5 KΩ	t _{on}	All	—	8	—	μs
Turn-Off Time	I _F = 30 mA, V _{CC} = 5 V, R _L = 2.5 KΩ	t _{off}	All	—	50	—	μs

Figure 1. Output Current vs. Input Current

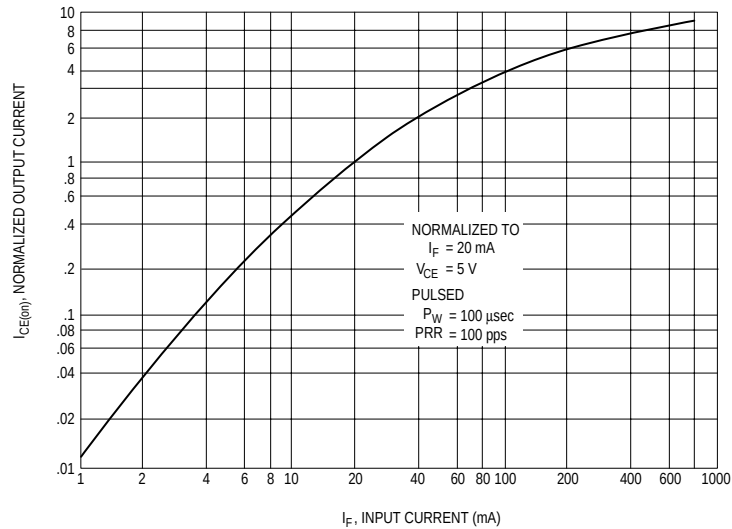


Figure 2. Output Current vs. Temperature

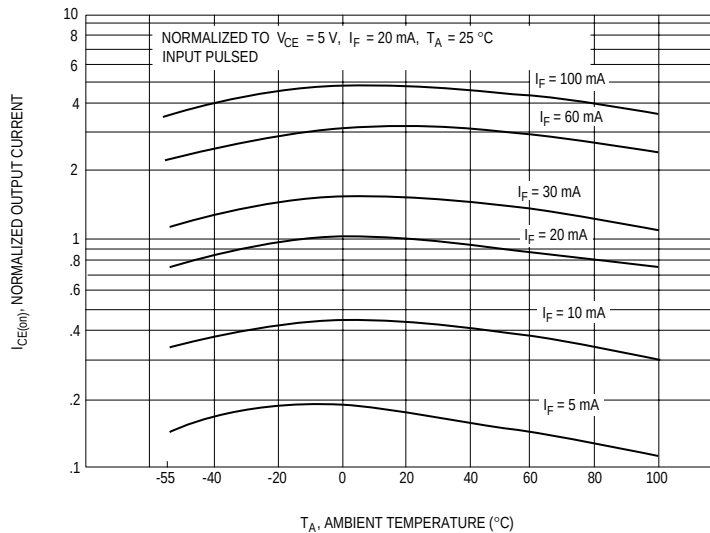


Figure 3. $V_{CE(SAT)}$ vs. Temperature

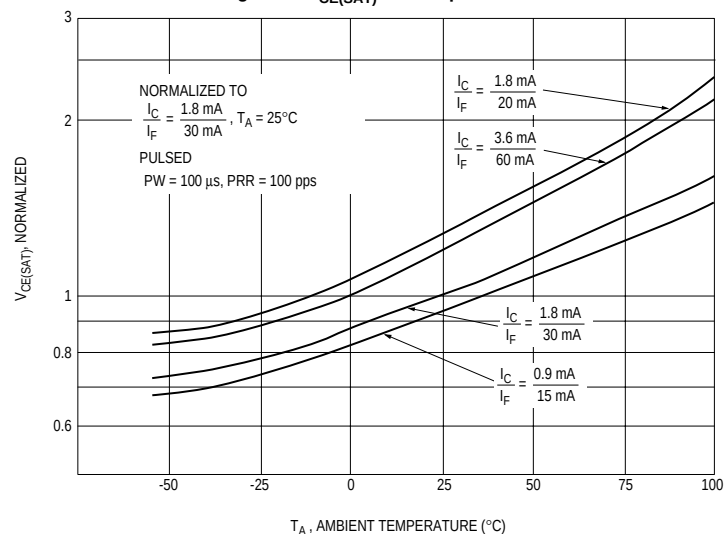


Figure 4. Leakage Current vs. Temperature

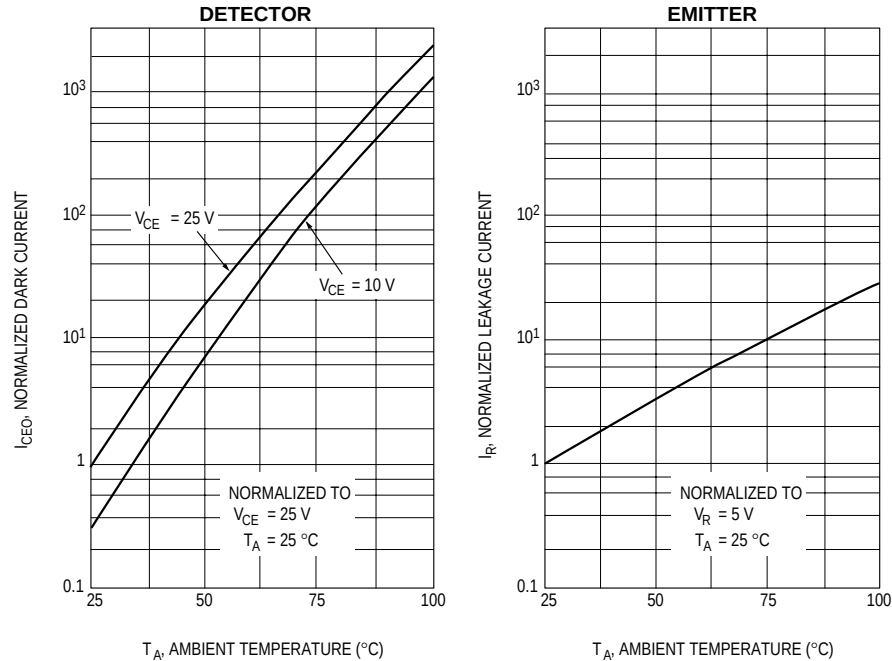


Figure 5. Switching Speed vs. R_L

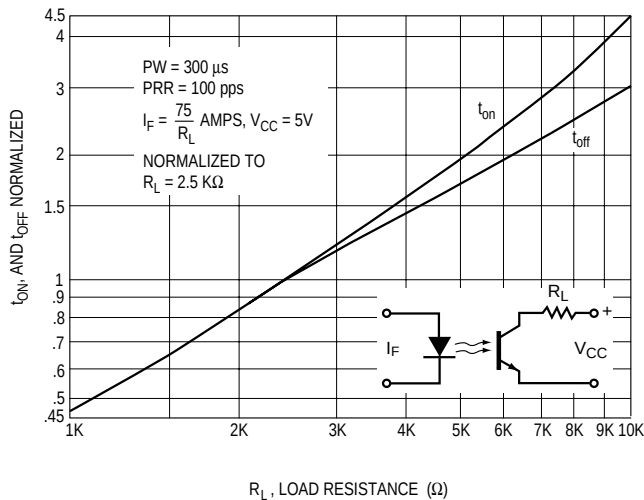
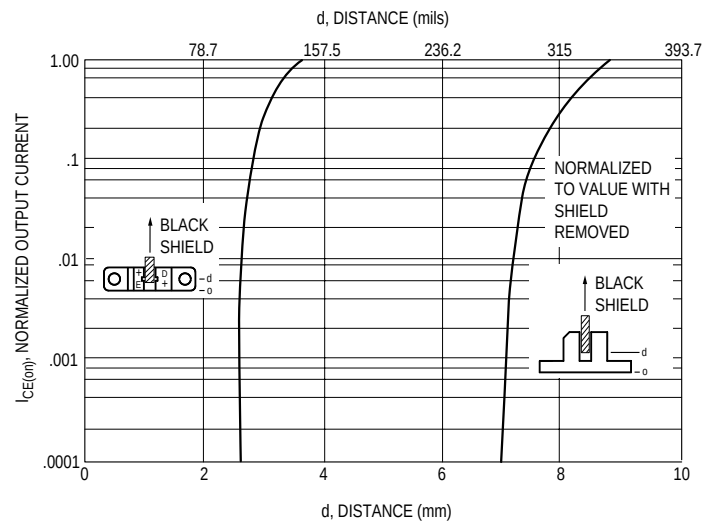


Figure 6. Output Current vs. Distance



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